

Stock Return Analysis for Axsome Therapeutics

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This report analyzes the historical stock returns of Axsome Therapeutics Inc, abbreviated as AXSM on the Nasdaq (NYSE).

1. About Axsome Therapeutics

Axsome Therapeutics was founded in 2012 by Herriot Tabuteau, a physician that wanted more expansive drug development for Central Nervous System (CNS) disorders like Alzheimer's, Depression, and even smoking cessation. They look to expand innovation in the CNS treatment field, to provide additional options for patients and physicians, making it's potential impact on the entire human race (as many individuals face issues with said CNS disorders) very high.

2. Market Capitalization

```
#loading necessary packages
library("xts")
library("quantmod")
library("zoo")
library("knitr")

#stock prices/indices from Yahoo Finance
axsm_price<- getSymbols(
  Symbols="AXSM",
  src="yahoo",
  auto.assign=FALSE,
  return.class="xts"
)

acad_price<- getSymbols(
  Symbols="ACAD",
  src="yahoo",
  auto.assign=FALSE,
  return.class="xts"
)

NASDAQ_index<- getSymbols(
  Symbols="^IXIC",
  src="yahoo",
  auto.assign=FALSE,
  return.class="xts"
```

```

)

#Extract adjusted share price to better reflect stock performance
axsm_price_adj<- Ad(axsm_price)
acad_price_adj<-Ad(acad_price)
NASDAQ_index_adj<-Ad(NASDAQ_index)

#Outstanding shares as of October 2025
axsm_n_shares<-49900000
acad_n_shares<-168710000

#Market Cap
axsm_marketcap<- axsm_price_adj*axsm_n_shares
acad_marketcap<-acad_price_adj*acad_n_shares

```

Axsome Therapeutics went public in November of 2015, listed on Nasdaq as AXSM. As of October 3, 2025, AXSM has a market capitalization of \$6.03 billion.¹ This makes AXSM a Mid-Cap company, with \$2-\$10 billion cap being the range for Mid-Cap companies. For comparison, ACADIA Pharmaceuticals has a market cap of \$3.68 billion.

```

plot.zoo(axsm_marketcap/10^9,
         plot.type="single",
         main="Figure 1: Axsome Therapeutics' Market Capitalization Over Time",
         lwd=1,
         xlab="",
         ylab="Billion USD",
         family="serif"
)

```

¹Market capitalization is calculated by multiplying current stock value by the total number of outstanding shares.

Figure 1: Axsome Therapeutics' Market Capitalization Over Time

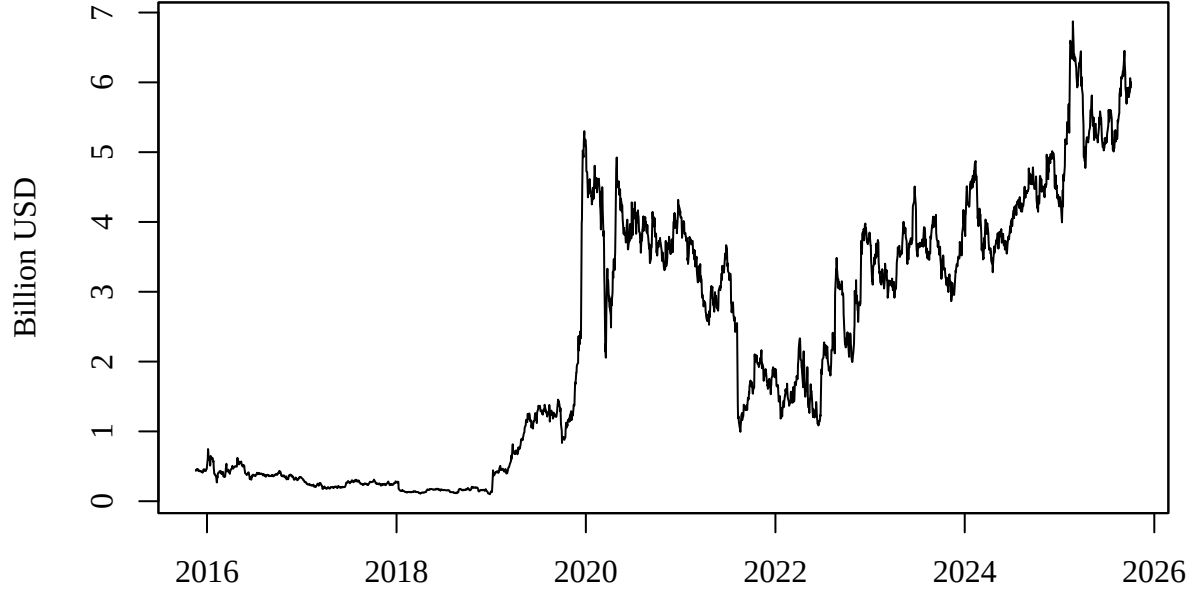


Figure 1 plots Axsome Therapeutics' Market Cap over time, reaching a peak of \$6.87 billion on February 21, 2025, dropping by ~\$0.87 billion in the eight months til now (October 2025). The drop can be attributed to a number of events in that time frame, including the FDA's refusal to file for the New Drug Application for AXS-14, leading to the requirement of more clinical trials, as well as allegations of inflating stock value, decreasing investor confidence in the stock.

3. A Visual Analysis of Stock Returns

When it comes to stock returns, investors focus on the key metrics, mean and standard deviation (or volatility) of monthly returns. The mean represents average return of the stock, while standard deviation quantifies the predictability of said returns. Returns on stock are defined as the measure of profit or loss from a stock investment; with that being said, log percentage return can be calculated as below.²

$$r_t = 100 \times (\ln(P_t) - \ln(P_{t-1}))\%$$

While the above equation calculates between monthly returns, an even more useful metric to track the performance of a company over a **longer** period of time is year-over-year (YOY) returns, essentially removing small fluctuations from the big picture. Using monthly prices, YoY returns can be quantified using the log difference as below.

$$r_t^{YoY} = 100 \times (\ln(P_t) - \ln(P_{t-12}))\% = \sum_{i=0}^{11} r_{t-i}$$

²Logarithmic returns are better than arithmetic ones because of their time-additive capabilities, while arithmetic returns are more suited for single, discrete time periods.

```

#Merge data & omit missing data points
Prices<-merge(axsm_price_adj,
              acad_price_adj,
              NASDAQ_index_adj)
Prices<-na.omit(Prices)

#Aggregation of Daily to monthly frequency
Prices_monthly<-apply.monthly(Prices,FUN=last)

#Returns and YoY Returns
Returns<- 100*diff.xts(Prices_monthly,
                       log=TRUE,
                       na.pad=FALSE,
                       lag=1)

yoy_Returns<-100*diff.xts(Prices_monthly,
                          log=TRUE,
                          na.pad=FALSE,
                          lag=12)

#Plotting YoY Returns
plot.zoo(yoy_Returns,
         plot.type="single",
         xlab="",
         ylab="%",
         main="Figure 2: Year-Over-Year Stock Returns",
         family="serif",
         col=c(5,2,1),
         lwd=c(3.5,1.5,1.5),
         lty=c(1,2,1)
        )
legend("topleft", legend = c("AXSM", "ACAD", "NASDAQ"),
      col = c(5, 2, 1), lwd = c(3.5, 1.5, 1.5), lty = c(1, 2, 1), bty = "n")

```

Figure 2: Year-Over-Year Stock Returns

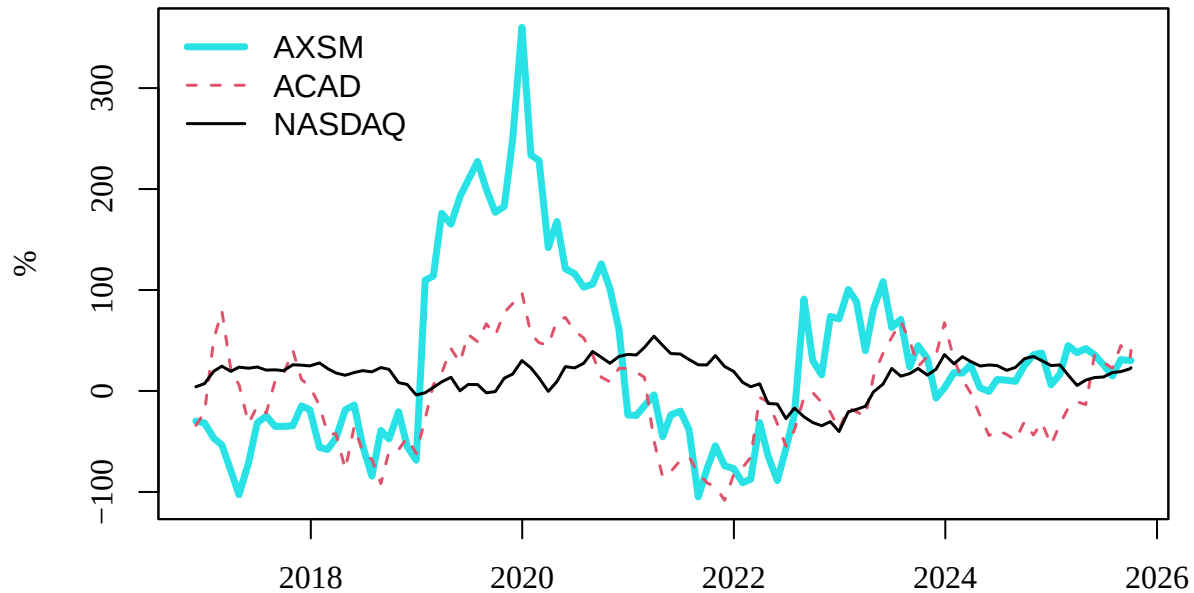


Figure 2 plots the different YoY returns for AXSM and ACAD vs the NASDAQ index for a ‘benchmark’. AXSM exhibits a much higher volatility, showing spikes of more than 300% in 2020, compared to ACAD and even NASDAQ in the same time frame. AXSM also showed an extreme drop in contrast to its ACAD counterpart, likely due to its lack of successful clinical trials; people were hesitant to invest in a company that hadn’t had successful trials at the time. Various rapid fluctuations in YoY stock returns can be attributed to clinical successes and setbacks at that specific time. It is also important to note the various allegations of stock value inflation and how that may have affected stock prices as well.

Although a visual representation is a valid way to see the general direction in which data takes us, it doesn’t give us exacts. For a more accurate and quantifiable comparison between the two companies, sample statistics were calculated.

4. Statistical Analysis of Stock Returns

4a. Overall Trend (Mean, Volatility, Min, Max, etc.)

```
# Assemble key statistics in a data frame
stats_dataframe<-rbind("Mean" = apply>Returns, 2, mean),
                      "Volatility (Standard Deviation)" = apply>Returns, 2, sd),
                      "Minimum Return" = apply>Returns, 2, min),
                      "Maximum Return" = apply>Returns, 2, max),
                      "Autocorrelation" = apply>Returns, 2, function(x) cor(x, lag.xts(x, 1),
use = "complete.obs")),
                      "Correlation to Nasdaq Index" = cor>Returns)[,3],
                      "Value" = as.numeric(tail(coredata(Prices), 1) / head(coredata(Prices), 1)))
value_label <- paste("Value of a $1 Investment in", format(start>Returns), "%b %Y"),
                  "by", format(end>Returns), "%B %Y"))
rownames(stats_dataframe)[rownames(stats_dataframe) == "Value"] <- value_label
colnames(stats_dataframe) <- c("AXSM", "ACAD", "NASDAQ")

# Define the time range for the statistics
time_interval <- format(x = range(index>Returns)), format = "%b %Y")
# Create table title

stats_title <- paste("Monthly $\\%-Return Statistics:",
paste(time_interval, collapse = " to "))

# Load knitr for table rendering
library("knitr")

# Format table with rounded statistics
kable(x = round(stats_dataframe, 2),
caption = stats_title,
format = "simple")
```

Table 1: Monthly %-Return Statistics: Dec 2015 to Oct 2025

	AXSM	ACAD	NASDAQ
Mean	2.19	-0.47	1.26
Volatility (Standard Deviation)	24.29	16.67	5.27
Minimum Return	-69.31	-64.08	-14.23
Maximum Return	109.03	48.24	14.36
Autocorrelation	-0.05	0.09	-0.05
Correlation to Nasdaq Index	0.19	0.31	1.00
Value of a \$1 Investment in Dec 2015 by October 2025	13.76	0.61	4.49

Table 1 presents the statistics from the monthly returns of AXSM, ACAD, and the NASDAQ index from November 2015 to October 2025.

AXSM shows erratic behavior when compared to ACAD and the NASDAQ benchmark, exhibiting the highest sd (volatility), as well as the lowest low (-69.31%) and highest high (109.03) on returns. This suggests that AXSM is a high-risk, high-reward stock for investors. This could be explained by a number of reasons: AXSM's young age as a company in conjunction with its mixed clinical trials make it easy for investors to jump ship at early signs of distress. At the same time, AXSM's potential to forever change the CNS

treatment industry attracts speculative buying with previously mentioned stock value inflation allegations explain the high volatility of AXSM.

This is further supported by each stock's correlation to the NASDAQ; AXSM having a lower correlation to NASDAQ than ACAD further suggests the volatile nature of AXSM, as the movement in ACAD stock value correlated to movements in the general market more so than AXSM. Both companies having a relatively low correlation to the NASDAQ may show the underlying volatile nature of pharmaceuticals

Furthermore, by looking at the 'Value of a \$1 Investment in Dec 2015-Oct 2025' column, we can see that AXSM brought a whopping 1300% increase, while ACAD actually depreciated by 39%, further contributing to the 'high-risk, high-reward' sentiment. Because the ages of both company's stock differ by ~10 years, one can only tell where AXSM may go; it depends on their future clinical trials, their ability to file NDAs, and how outside eyes respond to such.

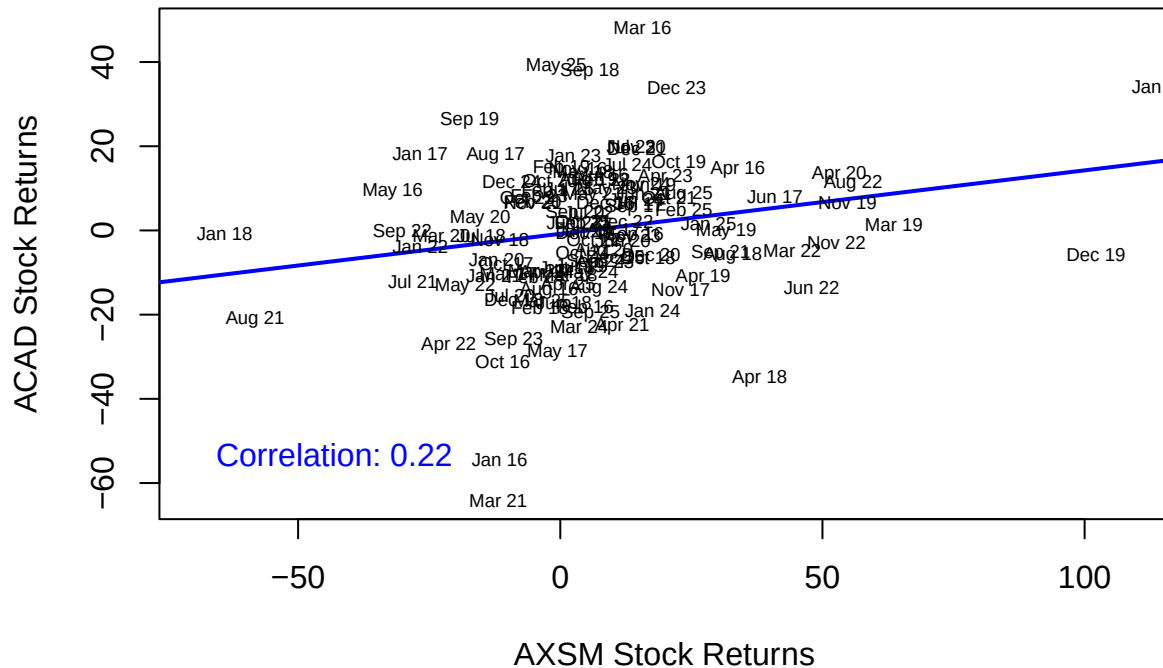
4b. Outlier Analysis

It is also important to look at potential outliers in the data, as they could have affected any of the above stats. Using a scatter plot to visualize the data via Figure 2 to ensure the viability of our data, ensuring that outliers did not have a play in higher volatility.

```
x_var<-Returns$AXSM.Adjusted
y_var<-Returns$ACAD.Adjusted
plot.default(x=x_var,
             y=y_var,
             xlab="AXSM Stock Returns",
             ylab="ACAD Stock Returns",
             main="Figure 3: Scatter Plot of Monthly Stock Returns",
             type="n")
abline(a=lm(y_var~x_var),
       col="blue",
       lwd=2)
corr_text<-round(x=cor>Returns)[1,2],
               digits=2)
text(x = x_var,
     y = y_var,
     labels = format(index(x_var), "%b %y"),
     adj = 0, cex = .6)

text(x = min(na.omit(x_var)), y = min(na.omit(y_var)) + 10,
     labels = paste("Correlation:", corr_text), pos = 4, col = "blue")
```

Figure 3: Scatter Plot of Monthly Stock Returns



The lower correlation value of 0.22 between AXSM and ACAD indicates that AXSM's stock fluctuations are likely not affected by ACAD's performance, and vice versa (This is further supported by the complete differences in sample statistics calculated above). With that being said, this further brings me to the conclusion that pharmaceutical companies' value is largely determined by their individual drug development endeavors; a company struggling in drug discovery is completely independent from another that may be finding success (theoretically shown through stock value), reflected in the 0.22 correlation.

#5. Conclusion

Axson Therapeutics' stock exhibits high volatility, making it a high-risk, high-reward investment despite its mid-sized market capitalization. This is due to the relatively volatile nature of pharmaceutical companies, as they depend on the success of their clinical trials to prove their value (in investors eyes) and solidify themselves as a 'trustable' company. Although the value of a \$1 investment in AXSM **far** trumps ACADs' growth, I believe AXSM's growth depends on their success in future clinical trials, as well as the dispelling of stock inflation allegations. Should they be able to overcome such problems, AXSM's growth up until current time suggests potential investment opportunities.